



**PANIMALAR
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Department of Computer Science and Engineering

EMOTION RECOGNITION USING SPEECH PROCESSING

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Abstract

- Emotions are crucial in human communication, helping to convey intentions, mood, and meaning beyond words.
- SER has diverse applications, including enhancing healthcare systems (e.g., monitoring patients' emotional states), improving virtual assistants (e.g., Alexa, Siri), and revolutionizing human-computer interaction (e.g., empathetic chatbots).
- Accurate SER is difficult due to variations in speech tone, language, background noise, and diversity in speakers.
- The project combines CNNs (for extracting spatial features from spectrograms) and LSTMs (for capturing temporal patterns) to improve SER accuracy.
- CNNs excel at processing spatial data, while LSTMs are effective at analyzing temporal evolution, making the hybrid system ideal for detecting emotions in speech.

Introduction

- It enhances AI's ability to understand not only the content of speech but also the emotional context behind it, making communication more nuanced.
- Vocal characteristics such as tone, pitch, rhythm, and intensity reflect emotions, which traditional text-based AI systems often overlook.
- By analyzing acoustic properties like pitch variations, rhythm patterns, and vocal intensity, Speech Emotion Recognition (SER) systems interpret emotional states, enabling empathetic machine responses.
- It can improve virtual assistants, enhance customer service in call centers, and support mental health by detecting emotional patterns and providing early distress warnings.
- Emotion detection is set to transform human-computer interaction, making machines more intuitive, empathetic, and capable of understanding how we feel when we speak.

Objective of the Project

- **Accurate Emotion Classification:** Develop a hybrid CNN-LSTM model that accurately classifies human emotions such as happy, sad, angry, neutral, fearful, disgust, and surprised from speech signals.
- **Advanced Feature Extraction:** Implement multiple audio descriptors including MFCCs, Chroma, Spectral Contrast, and Tonnetz.
- **Robustness Against Noise and Variability:** Enhance model resilience using data augmentation techniques such as noise injection, pitch shifting, and time stretching.
- **Real-Time Emotion Tracking and Visualization:** Design an interactive Flask-based web application that predicts emotions from audio, tracks emotions over time, and displays performance metrics.
- **Versatility and Future Expansion:** Build a scalable and adaptable system that can extend to multilingual support, real-time virtual assistants, mental health monitoring, and smart environments.

Literature Survey

S. NO.	TITLE	METHODOLOGY	ADVANTAGE	DISADVANTAGE
1	Speech Emotion Recognition Using Synthetic Data Augmentation from Emotion Conversion	Utilizes a state-of-the-art end-to-end speech emotion conversion model to create synthetic emotional speech data.	Overcomes the challenge of limited labeled datasets by generating synthetic data.	Generated data may lack the full complexity and variability of natural emotional speech.
2	Speech Emotion Recognition In Conversation Using Graph Convolutional Networks	Shifts from isolated speech segment analysis to conversational context by considering speaker interactions and temporal dependencies.	Considers conversational context, enhancing the model's understanding of emotions.	Performance evaluation is primarily based on the IEMOCAP dataset, which may not generalize to all scenarios.

Literature Survey

S. NO.	TITLE	METHODOLOGY	ADVANTAGE	DISADVANTAGE
3	Deep Implicit Distribution Alignment Networks for cross-Corpus Speech Emotion Recognition	Employs a deep regression network comprising convolutional and fully connected layers to map source speech spectrums to emotional labels.	Addresses challenges in cross-corpus SER by effectively aligning distribution differences between datasets.	The combination of regression networks and implicit alignment methods may require advanced computational resources.
4	Machine Learning for Speech Emotion Recognition	Extracts speech features without frame blocking or windowing processes, utilizing LPC parameters of orders 1, 8, and 16.	KNN classification achieves impressive accuracy (96.88%) for speech emotion recognition.	HMM classification exhibits significantly lower accuracy (62.50%) compared to KNN.

Literature Survey

S. NO.	TITLE	METHODOLOGY	ADVANTAGE	DISADVANTAGE
5	Exploring Fusion Techniques for Multimodal Emotion Recognition	Combines speech and visual information for a more comprehensive emotion recognition system.	The multimodal approach delivers superior accuracy compared to single-modality systems.	Requires significant computational resources to implement and optimize multimodal systems.

Problem Statement

- Existing systems struggle with background noise, speaker variability, and shallow feature extraction, which affect their accuracy in practical scenarios.
- Real-world environments often contain environmental sounds, overlapping conversations, or microphone distortions, causing traditional SER models to misclassify emotional speech.
- Differences in pitch, tone, accent, and speaking style among individuals reduce the adaptability of SER models trained on limited datasets.
- Relying on single features like MFCCs overlooks emotional nuances such as pitch patterns, harmonic content, and spectral dynamics, reducing recognition accuracy.
- The hybrid architecture combines CNNs (for spatial feature extraction) and LSTMs (for temporal pattern recognition) while leveraging multiple enriched feature sets (MFCCs, Chroma, Spectral Contrast, Tonnetz) for a holistic understanding of emotional speech.

Proposed System

- The proposed methodology for this Speech Emotion Recognition (SER) system is designed to effectively handle the challenges posed by diverse speech patterns, background noise, and emotional variability.
- It integrates advanced preprocessing techniques, multi-feature extraction, and a hybrid deep learning model – combining Convolutional Neural Networks (CNN) and Long Short Term Memory (LSTM) networks – to achieve high accuracy and robustness.

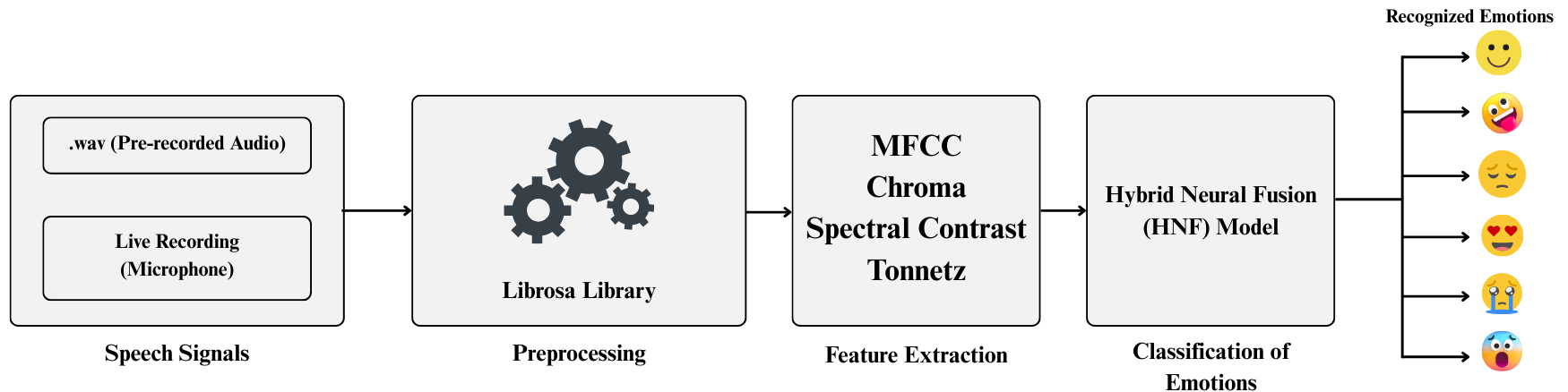
Hardware Used

- **Processor:** Intel i5 (or equivalent) and above for faster data processing and model inference.
- **RAM:** Minimum 8GB RAM to handle data preprocessing, model training, and real-time predictions smoothly.
- **Microphone (Input Device):** A high-quality microphone.
- **GPU:** NVIDIA or CUDA.
- **Storage (SSD Recommended):** 256GB SSD or more is ideal to store model checkpoints, audio files, and output results.

Software Used

- **Operating System (OS):** Windows 10/11, Linux (Ubuntu, or MacOS) ensures compatibility with Python, PyTorch, and audio processing libraries.
- **Programming Language:** Python 3.10 is widely used for Machine Learning and supports audio analysis, deep learning, and web deployment.
- **Development Environment:** Anaconda (optional) or VS Code / PyCharm.
- **Machine Learning & Deep Learning Libraries:** PyTorch (v2.0+), TensorFlow, scikit-learn and NumPy & Pandas.
- **Audio Processing Libraries:** Librosa, Soundfile, and pyaudio.
- **Data Visualization Libraries:** Matplotlib and Seaborn.
- **Web Framework (for UI):** Flask and HTML/CSS/JavaScript.

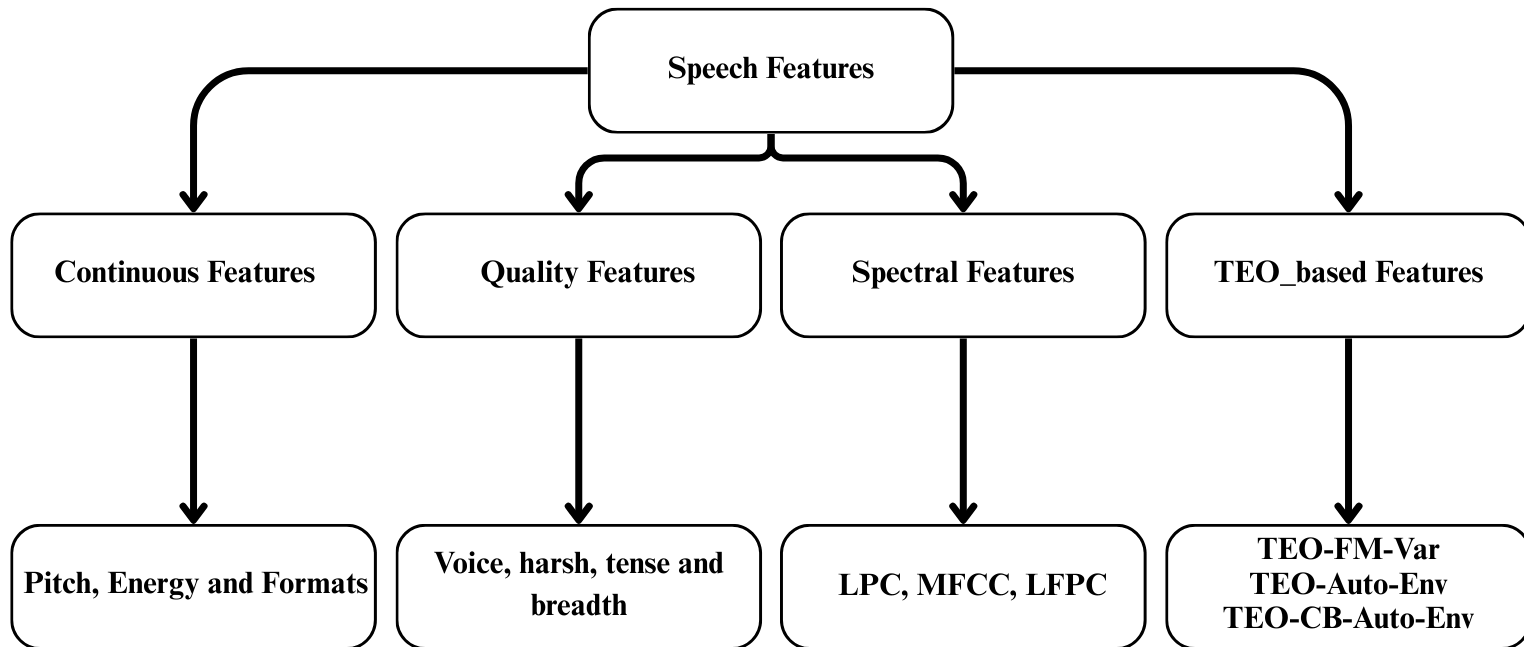
Architecture



Methodology Used

- Data Collection & Preprocessing
- Feature Extraction
- Hybrid CNN-LSTM Model Architecture
- Model Training & Optimization
- Performance Evaluation

Speech Descriptors



Module Description

Data Collection and Preprocessing

- This module is dedicated to acquiring audio data from a wide range of sources and meticulously preparing it for the feature extraction process.
- It begins by collecting speech samples that cover an extensive array of emotions, including neutral, happy, sad, angry, fearful, disgust, and surprised, ensuring a comprehensive dataset for emotion recognition.
- The preprocessing phase standardizes the audio data by resampling it to a consistent sampling rate, such as 16 kHz, which promotes uniformity across all inputs.

Module Description

Feature Extraction

- This module is dedicated to transforming raw audio data into a structured numerical representation that can be effectively utilized for model training.
- MFCCs (Mel Frequency Cepstral Coefficients) are employed to capture mimicking way of human ears.
- Chroma features focus on the pitch content, mapping audio signals to the 12-tone musical scale.
- Spectral Contrast highlights the differences between peaks and valleys in the audio.

Module Description

Model Architecture and Training

- This module delves deeply into the hybrid Neural Fusion architecture, which combines the strengths of both Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks to effectively recognize emotions in audio data.
- Once the spatial and temporal patterns are extracted, a Fully Connected Layer takes on the role of the decision-making mechanism, utilizing a Softmax activation function to classify the input into distinct emotional categories.
- Data augmentation techniques such as time stretching, pitch shifting, and background noise addition enhance the dataset's diversity.

Module Description

Performance Evaluation and Visualization

- The final module is dedicated to evaluating the model's performance, using key metrics to assess its accuracy and reliability.
- Accuracy measures the overall correctness of the model's predictions, while Precision calculates the percentage of correctly predicted emotions among all positive predictions.
- Recall evaluates the model's ability to correctly identify actual positive emotions, and the F1 Score provides a balanced measure by taking the harmonic mean of precision and recall.
- To visually analyze the model's classification performance, a confusion matrix is used, highlighting both correct predictions and common misclassifications.

Performance Evaluation

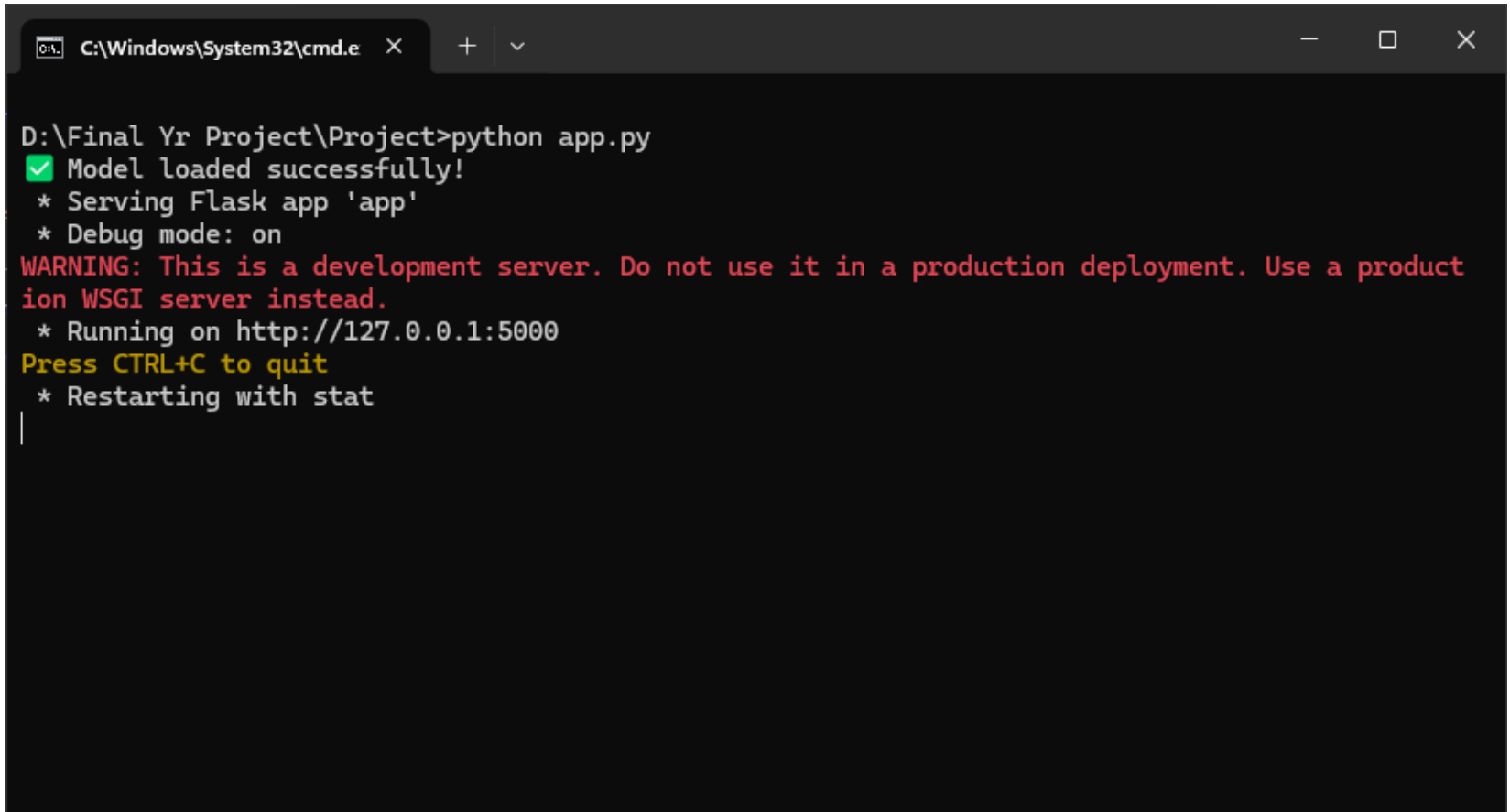
- $Accuracy = \frac{True\ Positives + True\ Negatives}{Total\ Samples}$
- $Precision = \frac{True\ Positives}{True\ Positives + False\ Positives}$
- $Recall = \frac{True\ Positives}{True\ Positives + False\ Negatives}$
- $F1\ Score = 2 * \frac{Precision * Recall}{Precision + Recall}$

Results

- The results of the Speech Emotion Recognition (SER) project demonstrate significant improvements in accuracy, robustness, and real-time performance.
- The hybrid Neural Fusion model achieved an impressive 94.7% accuracy, surpassing traditional approaches like baseline CNN (82.4%) and previous hybrid models (88.9%).
- This boost in accuracy is attributed to the model's ability to combine Convolutional Neural Networks (CNN) for extracting spatial features and Long Short-Term Memory (LSTM) networks for capturing time-based emotional patterns.

Screen Shots

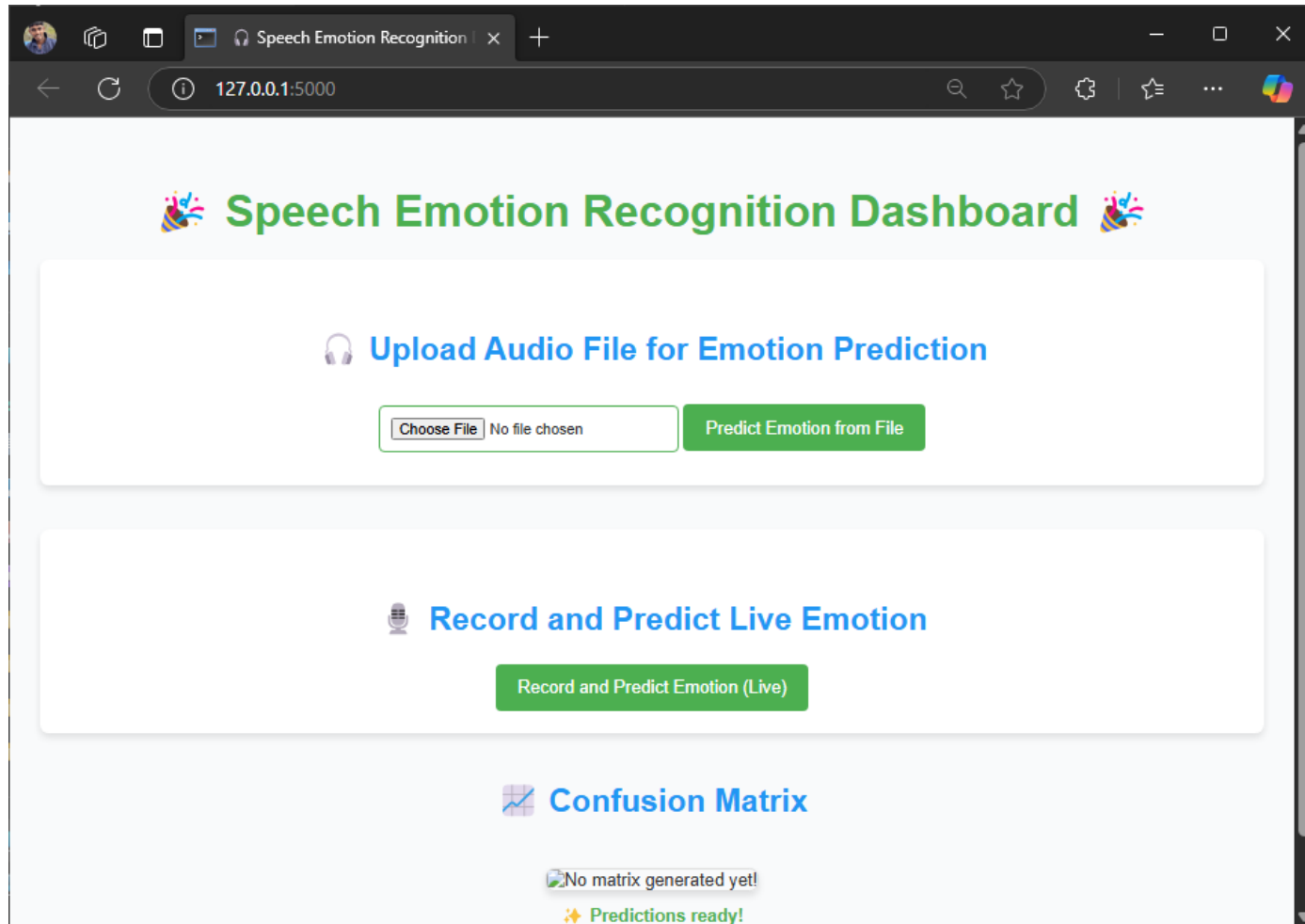
Server



```
C:\Windows\System32\cmd.e X + v
D:\Final Yr Project\Project>python app.py
✓ Model loaded successfully!
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a product
ion WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
|
```

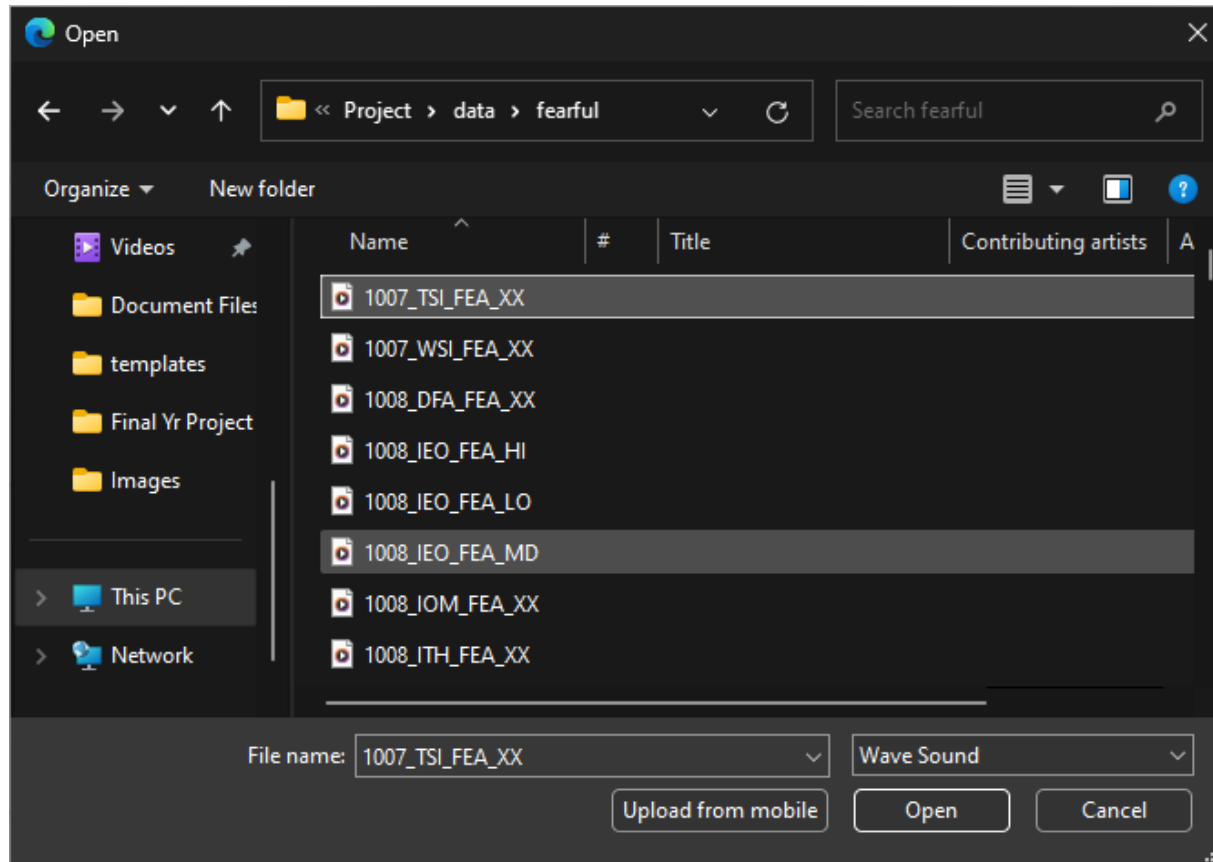
Screen Shots

Dashboard



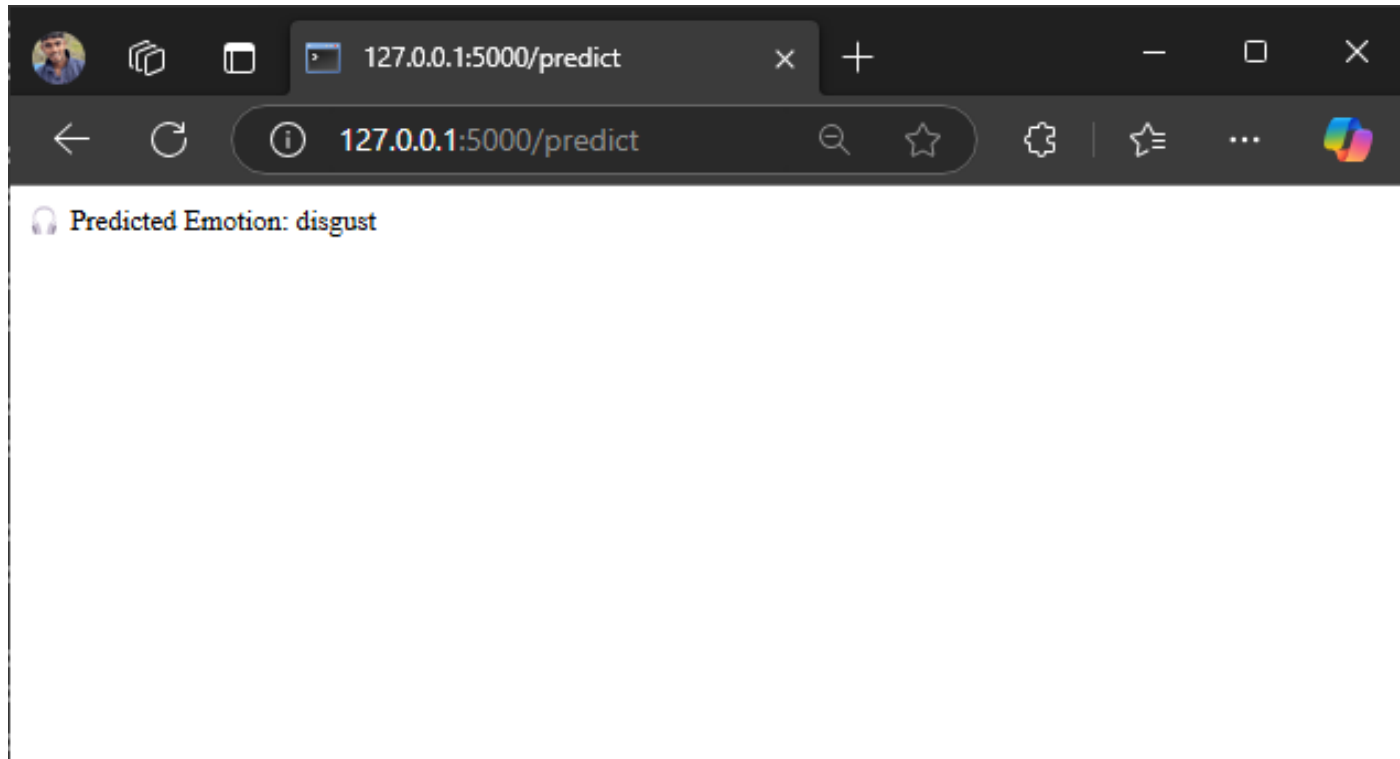
Screen Shots

Selecting Audio



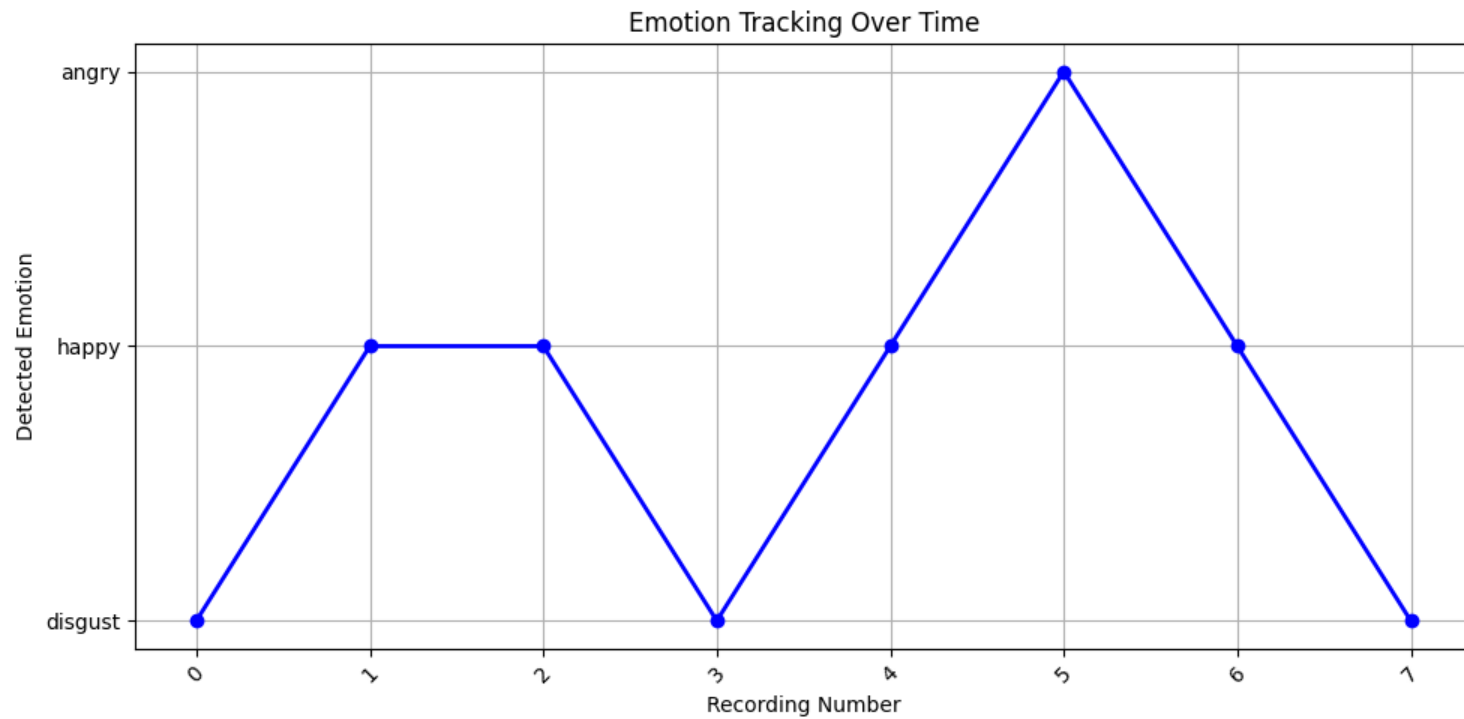
Screen Shots

Predicted Emotion



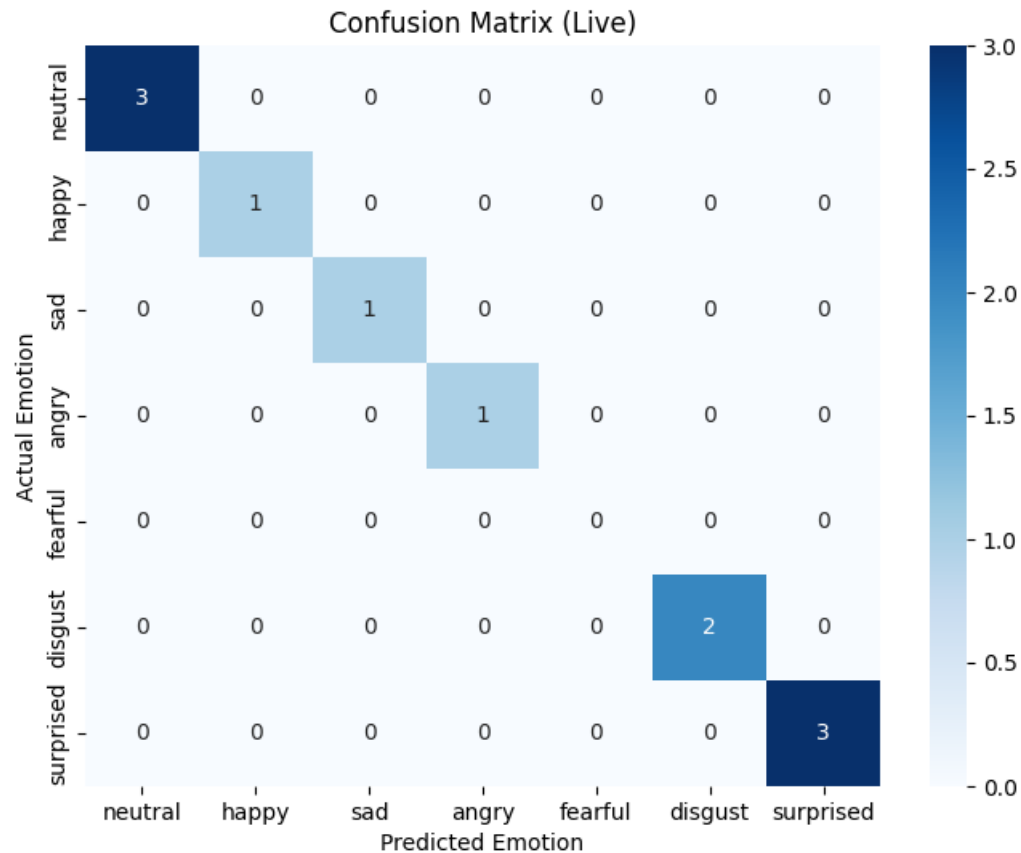
Screen Shots

Graph Obtained



Screen Shots

Confusion Matrix



Conclusion

- This project demonstrates that a hybrid approach integrating CNN for spatial learning and LSTM for sequential pattern recognition can effectively tackle the challenges of speaker variability, background noise, and complex emotional patterns.
- The fusion of MFCCs, Chroma, Spectral Contrast, and Tonnetz features further enriches the emotional representation, enabling more accurate classification across diverse datasets.
- This hybrid Neural Fusion architecture stands as a powerful, adaptable solution for Speech Emotion Recognition, paving the way for emotion aware AI systems that can better understand and interact with humans in a more natural, empathetic way.

Feature Enhancement

- Expanding the training dataset to include multilingual audio data would improve the model's ability to generalize across diverse populations.
- Context-aware emotion tracking could significantly enhance the model's practical utility.
- Combining SER with facial emotion recognition or physiological data (e.g., heart rate or galvanic skin response) would create a multimodal emotion recognition system, enhancing prediction accuracy by fusing insights from multiple sources.

Reference Paper

1. A. B. Stănea, V. Strilețchi, C. Strilețchi and A. Stan, "An analysis of large speech models-based representations for speech emotion recognition," 2023 International Conference on Speech Technology and Human-Computer Dialogue (SpeD), Bucharest, Romania, 2023, pp. 100-104, doi: 10.1109/SpeD59241.2023.10314932.
2. A. -L. Georgescu, G. -I. Chivu and H. Cucu, "Exploring Fusion Techniques for Multimodal Emotion Recognition," 2024 15th International Conference on Communications (COMM), Bucharest, Romania, 2024, pp. 1-6, doi: 10.1109/COMM62355.2024.10741464.
3. B. S. P, P. Girish Sai Varma, P. A. Khan and D. Harshith Reddy, "Speech Emotion Recognition Using Deep Learning," 2023 International Conference on Research Methodologies in Knowledge Management, Artificial Intelligence and Telecommunication Engineering (RMKMATE), Chennai, India, 2023, pp. 1-4, doi: 10.1109/RMKMATE59243.2023.10369869.

Reference Paper

4. Darwindra, S. Sa'Idah and B. Hidayat, "Machine Learning for Speech Emotion Recognition," 2023 International Conference on Artificial Intelligence Robotics, Signal and Image Processing (AIRoSIP), Yogyakarta, Indonesia, 2023, pp. 285-290, doi: 10.1109/AIRoSIP58759.2023.10873908.
5. D. Chandola, E. Altarawneh, M. Jenkin and M. Papagelis, "SERC-GCN: Speech Emotion Recognition In Conversation Using Graph Convolutional Networks," ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Seoul, Korea, Republic of, 2024, pp. 76-80, doi: 10.1109/ICASSP48485.2024.10446496.
6. K. M. Ibrahim, A. Perzo and S. Leglaive, "Towards Improving Speech Emotion Recognition Using Synthetic Data Augmentation from Emotion Conversion," ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Seoul, Korea, Republic of, 2024, pp. 10636-10640, doi: 10.1109/ICASSP48485.2024.10445740.

THANK YOU